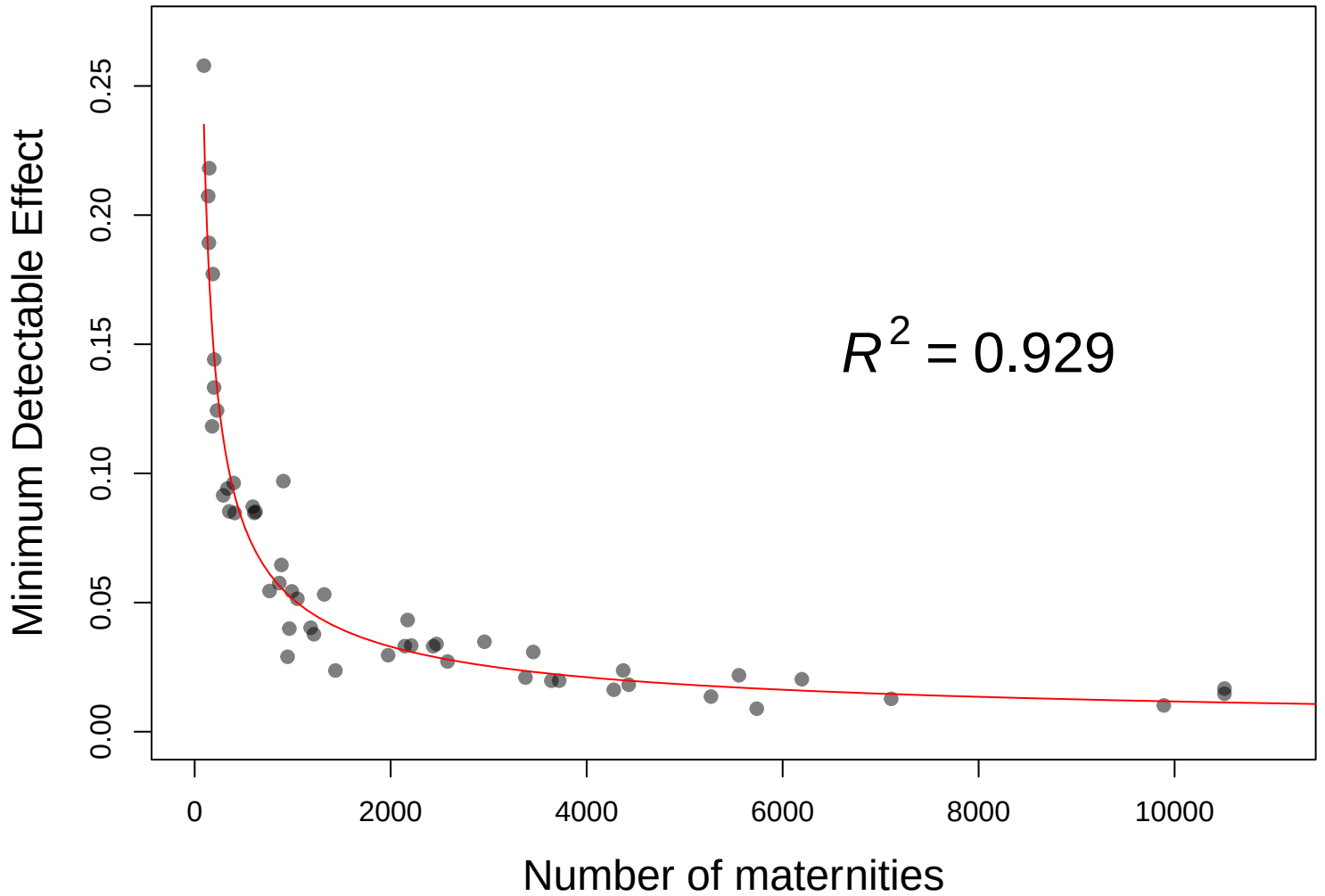
a)

Metric	R^2
records	0.74
maternities	0.929
paternities	0.924
full sibs	0.764
maternal sibs	0.906
maternal half sibs	0.704
paternal sibs	0.905
paternal half sibs	0.724
maternal grandmothers	0.574
maternal grandfathers	0.607
paternal grandmothers	0.591
paternal grandfathers	0.565
maximum pedigree depth	0.142
founders	0.216
mean maternal sibship size	0.546
mean paternal sibship size	0.495
non−zero F	0.262
F > 0.125	0.187
mean pairwise relatedness	−0.01
pairwise relatedness>=0.125	−0.004
pairwise relatedness>=0.25	0.044
pairwise relatedness>=0.5	0.156
var pairwise relatedness	0.043

b)